Software Engineering Trial Examination

2 hours 30 minutes including 10 minutes reading time.

Total 80 marks

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Question 1

What is a key feature of progressive web apps (PWAs)? (1 mark)

- ○ Require a constant Internet connection
- ○ Can be installed like native apps
- ○ Only work in Internet Explorer
- ○ Do not use JavaScript

Choose one answer only

Question 2

Which language is typically used for client-side programming? (1 mark)

- ○ PHP
- ○ Python
- ○ JavaScript
- ○ SQL

Choose one answer only

Question 3

A common machine learning model for unsupervised machine learning is (1 mark)

- ○ Logistic regression
- ○ Linear regression
- ○ Neural network
- ○ K-Nearest Neighbour

Choose one answer only

Question 4

Match the definitions to the concepts (2 marks)

Term	**Definition**
1.decryption	A) Converts data into a fixed-length value that cannot be reversed
2.digital Signature	B) Security protocols that enable encrypted communication over a network
3.encryption	C) Data that has not been encrypted nor protected
4.hashing	D) Converts encrypted data back into a readable format
5.plain text	E) Converts readable data into a coded format to protect privacy
6.SSL/TLS	F) Used to verify the authenticity and integrity of a message

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Question 5

Which of the following describe the Agile software development approach? (1 mark)

- ☐ It produces code faster than the Waterfall approach.
- ☐ It uses strict progression of steps through its life cycle.
- ☐ It uses an iterative workflow.
- ☐ It uses a software program called 'Agile'.

Choose as many answers as apply

Question 6

List 3 reasons why a web development team might use version control in the development of a website. (2 marks)

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Question 7

Describe the difference between input validation and input sanitisation as defensive data input handling practices, providing an example of each. (2 marks)

Question 8

Explain 3 differences between front-end and back-end web programming. (3 marks)

Question 9

Tools and processes can be used to generate solutions for clients based on their requirements. Match the tools and processes given below to their definition. (2 marks)

Term	**Definition**
1.algorithm	A) A table which represents data items, their data types, format, size and validation rules if they apply.
2.data dictionary	B) A sequence of sketches or mock-ups to represent how users navigate through different screens
3.quality assurance	C) A set of processes or rules to be followed to solve problems.
4.source code	D) Checking if developed software meets requirements or specifications.
5.evaluation	E) A set of instructions written in a programming language to perform specific tasks.
6.storyboard	F) Performance of tasks to ensure that developed software continues to meet requirements.

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Question 10 (4 marks total)

There are many different algorithms available for encrypting data, each with different strengths and weaknesses.

Part A

In which software development step would a software engineering team need to decide on the specific algorithm they will use? (1 mark)

- ○ Requirements Definition
- ○ Design
- ○ Development
- ○ Integration

Part B (3 marks)

Hashing algorithms are used to create a unique representation of data.

They have many uses in encryption and security.

Identify 3 essential features of a hashing algorithm.

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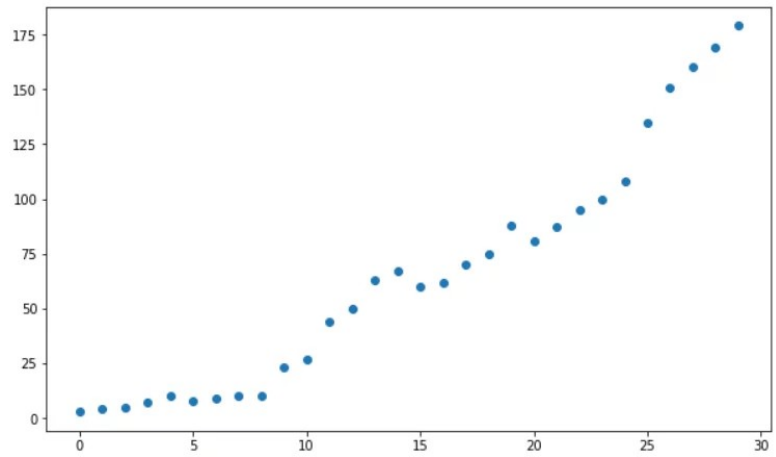
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Question 11 (6 marks total)

Which Machine Learning model is most appropriate for predicting new values based on the data given in the above graph?



Part A

Which Machine Learning model is most appropriate for predicting new values based on the data given in the above graph? (1 mark)

- ○ Decision tree
- ○ Polynomial regression
- ○ Linear regression
- ○ K Nearest Neighbour

Choose one answer only

Part B

Explain why a Neural Network would not be the best choice of model for the training data in the above graph and propose an alternative Machine Learning model. (5 marks)

Question 12

Consider the following scenario:

An Australian aerospace startup is developing software to control its rockets from launch on Earth to space and to landing on other moons and planets.

Each rocket must have:

- A rocket ID, e.g. R20
- Launch site, e.g. “Broome”
- Provider ID which identifies the company that made the payload, e.g. P01
- The dry mass of the rocket, e.g. 2500kg.
- The wet mass of the rocket, e.g. 5250.50kg
- Target orbit, which describes the orbit the payload is to enter
- A release() command which will open up the payload doors and unclamp the satellite from the cargo bay.

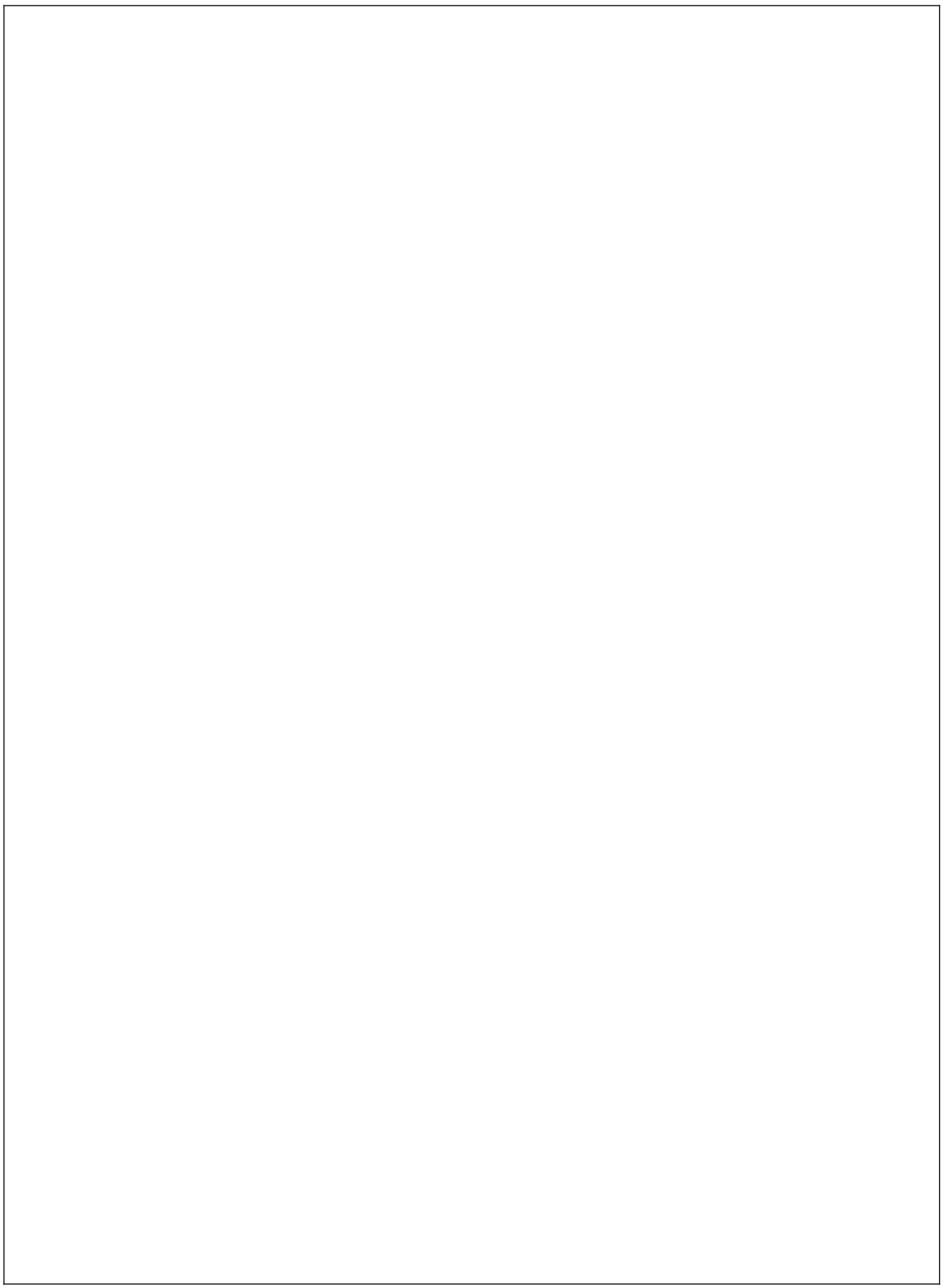
In addition, each satellite must have:

- Satellite purpose, which describes the mission of the satellite
- Satellite targets, which describes what countries, regions or areas of the Earth or other planets or moons are to be monitored by the satellite
- A signal() command, which is used to establish and maintain communications control with the provider as indicated by the Provider ID.

Answer the question on the following page

Question 12 continued

Create a class diagram for the rocket and satellite relationship described in the stimulus. (3 marks)



Question 13

A local music store is using a database table called albums: (3 marks)

albumID	albumname	format	albumprice
A001	Golden Hits	CD	18.50
A002	Live in Concert	Vinyl	22.00
A003	Acoustic Vibes	Digital	15.00
A004	Rock Legends	CD	20.00
A005	Chill Sessions	Vinyl	25.00

Write an SQL query to show the output below.

albumame	albumprice
Chill Sessions	25.00
Live in Concert	22.00

Question 14

Consider the following algorithm:

```
01  BEGIN checkLogin (username, password)
02      IF username = "" OR password = "" THEN
03          PRINT "Username or password empty."
04          RETURN False
05      ENDIF
06
07      length = 8
08      symbol = ['#', '$', '!', '^', '&', '[', ']']
09      symbol_present = False
10      # Minimum username/password length is given in a global variable
11      IF LENGTH(username) < length OR LENGTH(password) < length THEN
12          PRINT "Username or password is not long enough."
13          RETURN False
14      ENDIF
15
16      # Check if the first character (at index 0) of both username and password is a letter
17      IF (isalpha(username[0]) AND isalpha(password[0]) THEN
18          # Check for the presence of at least one symbol in either username or password
19          FOR j FROM 0 TO LENGTH(username) - 1
20              IF username[j] IN symbol THEN
21                  symbol_present = True
22              ENDIF
```

```
23        NEXT j
24        FOR j FROM 0 TO LENGTH(password) - 1
25            IF password[j] IN symbol THEN
26                symbol_present = True
27            ENDIF
28        NEXT j
29
30        IF NOT symbol_present THEN
31            PRINT "No symbol present in the username or password."
32            RETURN False
33        ENDIF
34    ELSE
35        PRINT "Username and password must start with a character."
36        RETURN False
37    ENDIF
38
39    // If all checks pass
40    RETURN True
41 END checkLogin
```

Answer the question on the following page

Question 14 ***(continued)***

Write Python code that implements as the function *has_symbol(username, password, symbols)* only the part of the above algorithm that checks whether either username or password contains a symbol from the given list of symbols and also prints a message if this is not satisfied. (4 marks)

Question 15

Consider the following scenario:

Click Shoe Warehouse is an online store for Australian-made shoes. They want to know which of their shoe sizes and styles sell the best in each season. They have prepared a spreadsheet with 10 years of monthly sales history, organised by style and size.

Explain how Machine Learning and Business Process Automation can help ‘Click Shoe Warehouse’ achieve its goal. (4 marks):

Question 16

An online service wants to reduce the chances of user accounts being compromised by dictionary attacks. In this type of attack, an attacker uses a pre-defined list (often referred to as a "dictionary") of words, phrases, or common passwords to guess the victim’s password

Describe TWO strategies they could implement to reduce this risk. (4 marks)

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Question 17

Web developers may use HTML and CSS or a CMS. Explain why an experienced web developer would choose to use HTML and CSS while a beginner would use a CMS. (4 marks)

Question 18

Assess positive and negative impacts of machine learning and artificial intelligence on production efficiency, waste and the environment. (5 marks)

Question 19 (6 marks total)

Consider the following scenario:

A library management system allows users to register online.

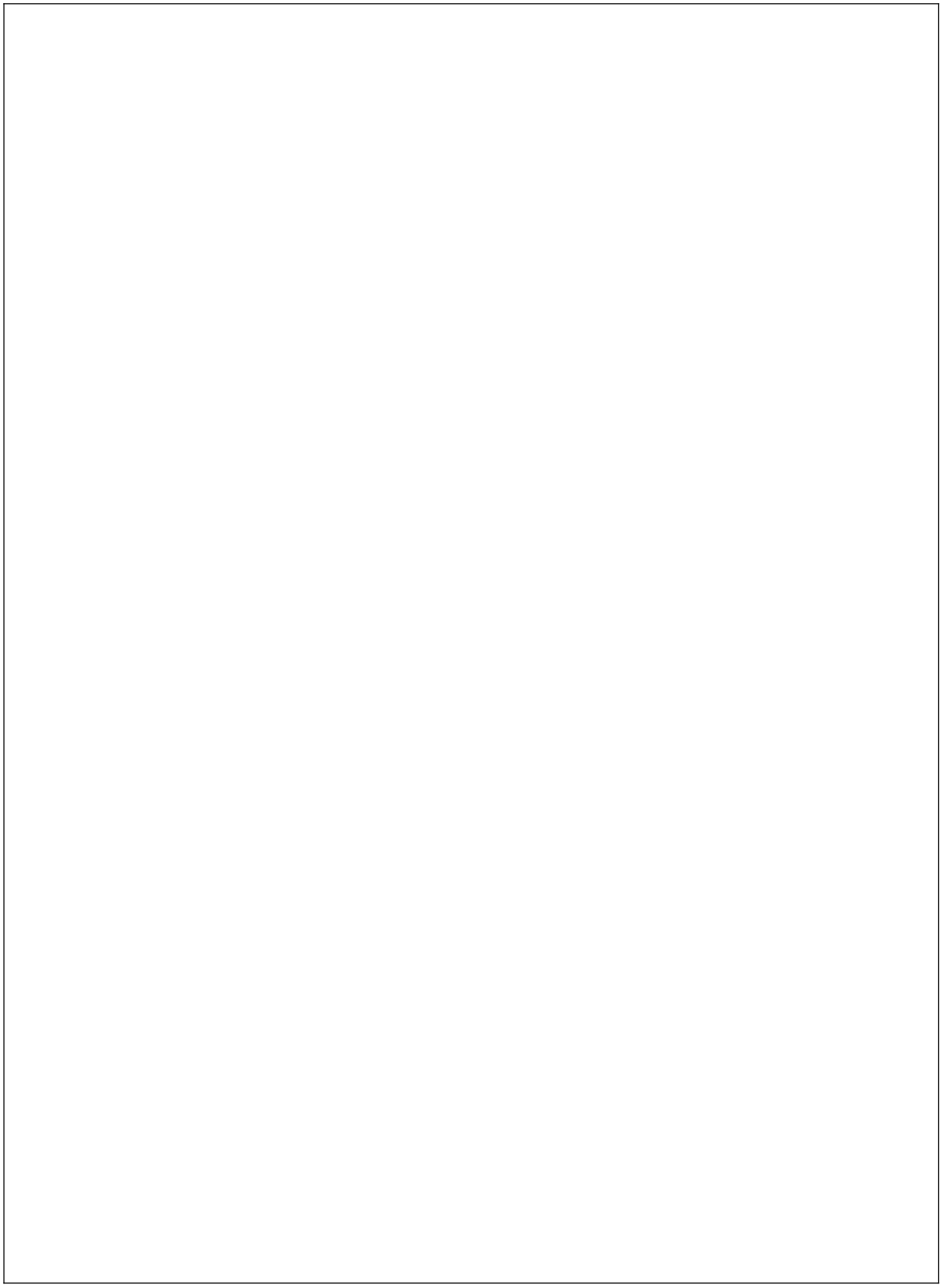
The registration form must collect user details and a secure PIN.

The PIN requirements are:

- Exactly 4 numeric digits
- No repeated digits (e.g. 1123 or 1231 is invalid)
- No letters nor special characters

Part A

Design a user interface for registration. Label all form fields clearly. (3 marks)



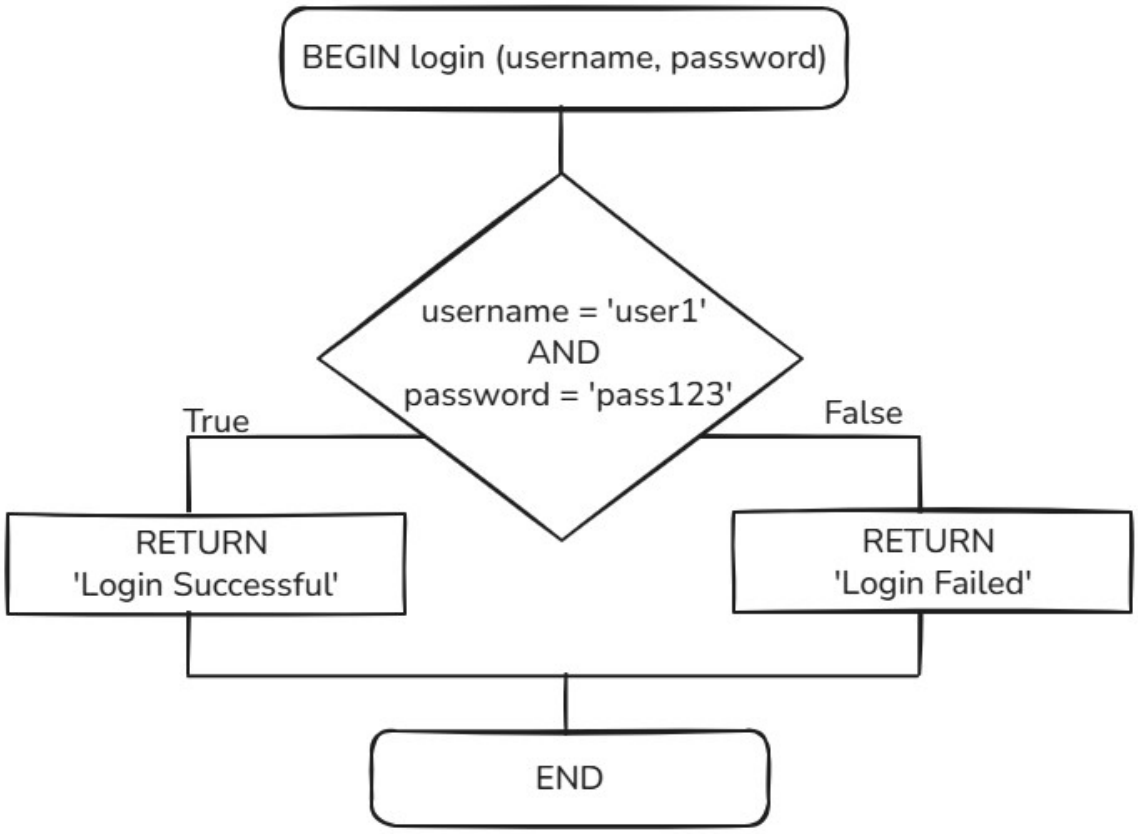
Part B

Explain how accessibility considerations improve the usability of web forms for a wide range of users, including those with disabilities. (3 marks)

Question 20

Describe the K Nearest Neighbour (KNN) algorithm for regression, predicting y for a new data point using a training data set of x and y values given the K value parameter of 3. (4 marks)

Question 21 (5 marks total)



Part A (1 Mark)

The algorithm flowchart shown describes a stub for a login function.

Implement this stub in python code.

Part B (2 marks)

One simple test has been provided below.

Provide at least TWO more tests that could be used, with a comment justifying the use of each one.

Test cases

```
IF login("user1", "pass123") == "Login successful" THEN
    PRINT "Test case 1 passed"
ELSE
    PRINT "Test case 1 failed"
```

Part C (2 marks)

You are tasked with writing an error message to return to users in situations where their login attempt fails.

This could happen when the password is entered incorrectly or when the username is incorrect.

What should you consider when creating these messages?

Question 22 (5 marks total)

Consider the following scenario:

An Australian toy manufacturing company has recently been established, with offices around the country. Toy manufacturing is subject to strict environmental, workplace health and safety, national and international regulatory frameworks, enforced by the agency AusToy.

Part A

Explain communications issues that may arise due to stakeholders being located all over the country. (3 marks)

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Part B

Compare and contrast Hybrid (also called WAgile) and Waterfall software development methodologies in developing software to control the manufacturing of their toys. (2 marks)

Question 23

A popular social media platform has recently suffered a cross-site scripting (XSS) attack, leading to significant user data breaches and a loss of public trust.

Discuss the ethical, legal, and technical ramifications for both users and the enterprise stemming from such a security breach. (8 marks)

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